

Graded Entity-Familiarity Readouts in Language Models: Polish Adaptation, Cross-Language Robustness, and Refusal Steering*

Grzegorz Brzezinka

Prosit AS

greg@prosit.no

Abstract

Can a language model estimate its familiarity with an entity before generating an answer? We study activations at the final prompt token in twelve instruction-tuned models from the Bielik, PLLuM, Gemma-4, and Qwen3 families, using a new dataset of 1,440 Polish entities spanning four domains and ten Wikipedia-pageview deciles, plus fabricated controls. Familiarity-probe scores separate real from fabricated entities in every family; in the Polish-adapted Bielik and PLLuM families they additionally track entity popularity (model-mean Spearman ρ 0.28–0.57, versus at most 0.11 in Gemma-4 and Qwen3), a pattern more strongly associated with Polish adaptation than with parameter count in this model sample. In a paired experiment on two families, probes retain 96–101% of within-language AUROC when the Polish question stem is replaced with an English one around unchanged entity names, showing robustness to prompt language in this setting. In Gemma-4-12B, the only model that natively refuses, adding a one-dimensional familiarity direction at a single layer moves refusal rates monotonically in both directions (0.24 to 1.00 on well-known entities; 0.73 to 0.00 on unknown ones). Finally, a calibrated familiarity probe is competitive among pre-generation abstention gates, although post-generation detectors better predict behavioral error on average. These results support a graded pre-generation entity-familiarity readout, and a separation between representational familiarity and the policy that converts it into abstention.

1 Introduction

When a language model is asked about an entity it has never encountered, it can either abstain or confabulate. Prior work shows that the model’s internal states contain information relevant to this choice (Kadavath et al., 2022; Azaria

and Mitchell, 2023; Orgad et al., 2025), and Ferrando et al. (2025) localized an entity-recognition mechanism in sparse-autoencoder space. Closest to our own work, Gottesman and Geva (2024) showed that supervised probes over an entity’s *pre-generation* hidden states predict graded QA accuracy. Most detection methods, however, score *generated answers* (Ding et al., 2025; Ettori et al., 2025; Su et al., 2024; Farquhar et al., 2024). A recent study of the Polish Bielik family (Brzezinka, 2026) instead measured the *prompt point* (the last token of the question, before any answer token exists) and found that unsupervised activation-dispersion metrics separate well-known from fabricated Polish entities at AUROC 0.95–1.00 across scales from 1.5B to 11B parameters, while the same signal says much less about whether the answer will be factually correct.

That result was binary (known vs. fabricated), single-family, single-language, and purely correlational. It therefore left open the questions that determine what the signal *is*:

- **RQ1 (language).** Is the signal about the entity or the question’s surface form? If a probe trained on Polish questions transfers to English questions about the same entities, the readout is at least robust to the question-stem language.
- **RQ2 (gradation).** Is familiarity a bit or a quantity? If the score tracks a continuous popularity axis, it behaves as a graded quantity, and the predecessor’s ceiling is a property of an easy contrast, not of the signal.
- **RQ3 (causality).** Does the model itself read this signal, or is the familiarity–refusal link merely correlational?
- **RQ4 (application).** Can the signal gate generation (abstain or route to retrieval *before* producing an answer) competitively with

*Preprint. Under review.

post-generation detectors, at a fraction of their cost?

We answer all four with one fixed one-forward-pass measurement protocol (§3), applied to a new popularity-graded dataset of Polish entities (§4) and twelve instruction-tuned models from four families spanning a Polish-pretraining axis, from Polish-native Bielek through PLLuM (Polish continual pretraining on Llama-3.1 and Mistral-NeMo bases) to the multilingual, non-Polish-centric Gemma-4 and Qwen3. Coverage varies by question: gradation (RQ2) uses all twelve models on the new dataset (v2) and gating (RQ4) the largest model per family on v2, while language transfer (RQ1; six models) and steering (RQ3; Gemma-4-12B) predate v2 and use the predecessor’s three-condition dataset (v1; §4).

Our contributions, in narrative order: (1) a QID-resolved, decile-stratified Polish entity dataset with fabricated anchors, token-length matched to the real distribution (§4); (2) evidence that the familiarity score is graded in the Polish-adapted families and that its gradation is more strongly associated with Polish adaptation than with scale in this model sample, with a behavioral mirror in answer correctness (§5); (3) zero-shot cross-language probe transfer at 96–101% of within-language performance in the two families tested, arguing against a large question-stem-language effect in this paired setting (§6); and (4) an intervention-plus-application study: a single rank-one familiarity direction steers Gemma-4-12B refusal monotonically in both directions ($0.24 \rightarrow 1.00$ / $0.73 \rightarrow 0.00$), extending Ferrando et al. (2025) with a dose–response characterization (§7), and a risk–coverage comparison against adapted post-generation baselines that positions the calibrated probe within the pre-generation gate class and quantifies the dissociation between representational familiarity and behavioral abstention policy (§8, §9).

2 Related Work

Internal-state hallucination detection. Supervised probes on hidden states detect falsehoods and hallucinations (Azaria and Mitchell, 2023; Orgad et al., 2025; Du et al., 2024; Chen et al., 2024; Sriramanan et al., 2024; Binkowski et al., 2025; Park et al., 2025); semantic entropy measures answer instability across samples (Kuhn et al., 2023; Farquhar et al., 2024; Kossen et al., 2024);

SelfCheckGPT is sampling-based and black-box (Manakul et al., 2023); unsupervised linear truth directions (Burns et al., 2023; Marks and Tegmark, 2024) and verbalized confidence (Lin et al., 2022; Tian et al., 2023) offer complementary readouts. Closest to our setting are question-only approaches: Kadavath et al. (2022) elicit self-knowledge behaviorally, Moreno Cencerrado et al. (2025) train question-side probes, Slobodkin et al. (2023) find answerability encoded before generation, Snyder et al. (2024) detect hallucinations from pre-generation artifacts, and Ferrando et al. (2025) identify entity-recognition features that causally modulate refusal in Gemma 2. Gottesman and Geva (2024) estimate entity knowledge from pre-generation hidden states with supervised probes that predict graded QA accuracy; this is the nearest precedent to our readout. We differ in the popularity-decile design with a fabricated anchor and in adding cross-family, cross-language, and interventional axes; of our two readouts, only dispersion is unsupervised and closed-form, while our best-performing probe is, like theirs, a supervised classifier. Obeso et al. (2025) carry entity-recognition probes into real-time detection. Linear representations are known to track pretraining term frequency (Merullo et al., 2025; Mozafari et al., 2026), and internal states reflect knowledge recall rather than truthfulness (Cheang et al., 2026), converging with our familiarity-not-truth framing. Recent single-generation detectors such as D²HScore (Ding et al., 2025), EigenTrack (Ettori et al., 2025), and MIND (Su et al., 2024) score the answer’s hidden states; we compare against documented adaptations of all three (§8).

Popularity and the long tail. Factual accuracy tracks entity popularity (Mallen et al., 2023) and pretraining co-occurrence (Kandpal et al., 2023). We turn this external axis into a probe of the *internal* signal: if familiarity is a readout of exposure, its score should be monotone in log-popularity.

Activation steering. Adding difference-of-means directions to the residual stream steers behavior (Turner et al., 2023; Rimskey et al., 2024); inference-time intervention steers truth-related directions (Li et al., 2023); refusal in particular is mediated by a single direction (Arditi et al., 2024), and conditional or surgical variants refine where and when to steer (Lee et al., 2025; Yang et al., 2025b). We apply this methodology not to jailbreaking but to the epistemic axis: does

moving along the familiarity direction move the model’s own abstention decision?

Polish evaluation. Mu-SHROOM (Vázquez et al., 2025) and MultiHaluDet (Alvi et al., 2026) cover Polish hallucination annotation; Bielik (Ociepa et al., 2025a,b) and PLLuM (Kocoń et al., 2025) are the Polish-adapted model families we study, alongside Gemma-4 (Gemma Team, Google DeepMind, 2026) and Qwen3 (Yang et al., 2025a); PLLuM’s 8B and 12B variants build on Llama-3.1 (Grattafiori et al., 2024) and Mistral-NeMo (Mistral AI, 2024).

3 Familiarity Readouts

All measurements are taken at the *prompt point*: the final token of a one-sentence Polish question (*Kim jest {entity}? Odpowiedz jednym zdaniem.* for people; *Czym jest ...* for places), in one forward pass, before generation. Following Brzezinka (2026) we use two signal classes plus a baseline: **dispersion** (unsupervised: per-layer inverse participation ratio and Shannon entropy over the post-SwiGLU MLP activation vector; best (metric, layer) cell selected on an anchor contrast, here top-decile-vs-fabricated, oriented so higher = more familiar), a **probe** (logistic regression on the residual-stream hidden state, layer selected by 5-fold CV, training-contrast scores always out-of-fold), and **first-token entropy**. Conventions: separability = $\max(\text{AUROC}, 1 - \text{AUROC})$; bootstrap 95% CIs; fixed seeds. Functionally analogous readouts appear in every family (on Gemma-4-12B the best layers sit in the same mid-depth relative band as Bielik, 0.64–0.71 on athletes, and the per-head picture is equally diffuse: rank consistency 0.57 vs. 0.56, 2 of 768 heads in all four domain top-20 sets), so one measurement recipe applies to every family without retuning. Because each readout is trained or selected per model, similar performance and layer placement do not establish that the families share an identical representation; cross-model transfer or representational-similarity tests would be needed for that stronger claim.

4 A Popularity-Graded Dataset

Why graded. The predecessor’s known-vs-fabricated contrast is at ceiling already at 1.5B, capping what scale comparisons can reveal. A pilot on its 336 real entities found the familiarity score is not a bit but a *graded popularity readout*:

Spearman correlation with log 12-month pl.wiki pageviews averaged $\rho = 0.75$ across 6 models $\times$ 4 domains, surviving within the well-known tier alone; pageviews and Wikidata sitelinks were interchangeable. This motivated sampling the popularity axis continuously.

Design. For each of four domains (athletes, cities, writers, musicians) we harvest a full candidate pool from Wikidata SPARQL (occupation- or class-filtered, Polish citizenship or location, Polish-Wikipedia article required), resolve every entity by QID (excluding disambiguation pages by construction) and stratify 300 real entities into ten deciles of log 12-month pl.wiki pageviews (30 per decile), plus 60 fabricated anchor entities per domain: morphologically valid invented Polish names, screened for non-existence by exact page lookup and full-text search, and selected so that real-vs-fabricated token-count AUROC stays near chance under both the smallest and largest Bielik tokenizers (achieved 0.50–0.57, target < 0.65); a post-hoc check confirms the screen generalizes to all four family tokenizers (pooled token-count AUROC 0.49–0.52, worst single domain cell 0.56). All sampling is deterministic (`random.Random(0)`); the totals are 1,200 real + 240 fabricated = 1,440 entities. The steering and language-transfer experiments (§6, §7) predate v2 and use the predecessor’s three-condition dataset (42 well-known / 42 obscure-real / 42 fabricated entities per domain), which we call v1.

Models. Twelve instruction-tuned checkpoints: Bielik v3.0 (1.5B, 4.5B, Minitron-7B (Kinas et al., 2026), 11B), Gemma-4 (E4B, 12B), PLLuM (4B; Llama-PLLuM-8B; 12B, Mistral-NeMo-based; all instruct-2512), and Qwen3 (1.7B, 4B, 14B, with thinking disabled). The families span the axis we care about: Polish-native pretraining (Bielik), Polish continual pretraining over non-Polish bases (PLLuM), and multilingual non-Polish-centric training (Gemma-4, Qwen3).

5 RQ2: Familiarity Is Graded and Aligned with Polish Adaptation

The signal is graded. Figure 1 shows per-decile familiarity curves; Table 1 and Figure 2 summarize gradation strength. Classifying each of the 48 model $\times$ domain decile-vs-fabricated AUROC curves by their trend (Spearman ρ of the ten AU-

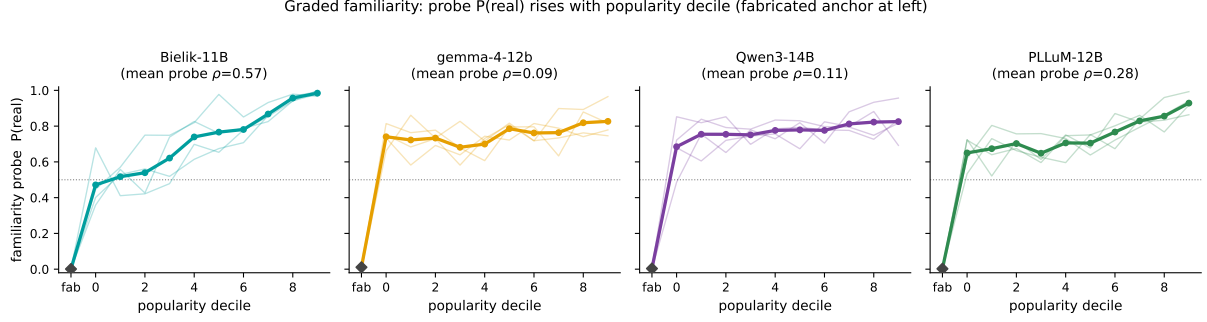


Figure 1: Per-decade familiarity curves on dataset v2 (largest model per family; probe $P(\text{real})$; fabricated anchor at left, then popularity deciles 0–9; thin lines are individual domains). Bielik-11B rises through the entire popularity range (mean $\rho = 0.57$); Gemma-4-12B and Qwen3-14B separate the fabricated anchor from real entities but vary little across popularity deciles ($\rho = 0.09/0.11$); Polish-adapted PLLuM-12B is intermediate ($\rho = 0.28$).

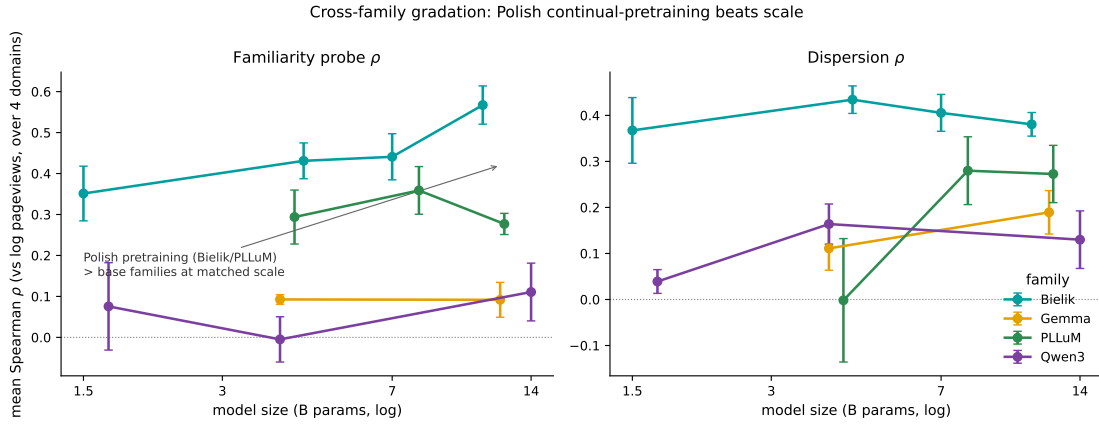


Figure 2: Gradation strength (Spearman ρ of familiarity score vs. log pageviews among real entities, mean over four domains) by model size and family. Qwen3 stays near zero from 1.7B to 14B, while the Polish-adapted models (Bielik, PLLuM) show stronger gradation at every matched size.

ROCs vs. decile index: rising $\rho \geq 0.5$, falling $\rho \leq -0.5$, else flat), 27 of 48 are rising and none crosses the falling threshold. This trichotomy is coarse (ten deciles, an arbitrary ρ cutoff), so we read it as a descriptive summary rather than primary evidence; the per-cell correlations with uncertainty are in the released artifacts.

Gradation aligns with Polish adaptation, not scale, in this sample. Within Bielik, the probe’s mean gradation grows from $\rho = 0.351$ (1.5B) to 0.567 (11B). But scale alone does not produce gradation: Qwen3 moves from 0.076 to 0.111 between 1.7B and 14B, and Gemma-4-12B sits at 0.092. The most informative observational comparison is PLLuM: after continual pretraining on Polish text, the Llama-3.1-based 8B reaches $\rho = 0.359$ and the Mistral-NeMo-based 12B 0.277, exceeding the matched-size non-Polish models by 0.17–0.27 absolute (Gemma-4-12B 0.092, Qwen3-14B 0.111). We now have

controlled before/after evidence for both PLLuM architectures: each base checkpoint grades well below its Polish-adapted descendant at matched architecture, tokenizer, and scale. Mistral-NeMo-12B (base of PLLuM-12B) scores mean prompt-point $\rho = 0.102$ against PLLuM-12B’s 0.277, and Llama-3.1-8B (base of PLLuM-8B) scores 0.193 against PLLuM-8B’s 0.359 – probe lifts of 0.175 and 0.166 from Polish continual pretraining alone (dispersion lifts 0.094 and 0.223). The Mistral-NeMo base sits at the non-Polish level, so the adaptation, not the architecture, produces the gradation. An exact permutation test over the twelve model-level mean ρ values ($\binom{12}{7} = 792$ family-label assignments) puts the Polish-vs-non-Polish gap at 0.315 for the probe (Cohen’s $d \approx 4.1$; $p = 0.0013$, the exact null’s resolution floor) and 0.179 for dispersion ($p = 0.019$); the partial correlation with Polish adaptation controlling for log-size is 0.905. Because checkpoints within a family

Family	Model	B	disp. ρ	probe ρ
Bielik	1.5B	1.5	0.367	0.351
Bielik	4.5B	4.5	0.434	0.431
Bielik	Minitron-7B	7.0	0.405	0.441
Bielik	11B	11.0	0.381	0.567
Gemma-4	E4B	4.0	0.111	0.093
Gemma-4	12B	12.0	0.189	0.092
PLLuM	4B	4.3	-0.002	0.294
PLLuM	Llama-8B	8.0	0.280	0.359
PLLuM	12B	12.25	0.273	0.277
Qwen3	1.7B	1.7	0.039	0.076
Qwen3	4B	4.0	0.164	-0.005
Qwen3	14B	14.0	0.130	0.111

Table 1: Gradation (Spearman ρ of familiarity score vs. $\log_{10}(\text{pageviews} + 1)$ among real entities; mean over four domains; bootstrap CIs per cell in the released artifacts). Dispersion = best-layer metric at the top-decile-vs-fabricated anchor; probe = $P(\text{real})$ from a top-3-deciles-vs-fabricated logistic probe. Caveat: dispersion’s cell selection uses the anchor AUROC, which couples it to ρ (Spearman 0.795 across the 48 cells), a mild optimistic bias; the probe is exempt, as its scores are out-of-fold.

share corpora, tokenizers, and training pipelines, they are not independent replications: the effective sample is closer to four families than twelve models, so we emphasize the effect sizes over the checkpoint-level p -values.

Behavioral mirror. We next test whether the internal gradient corresponds to anything behavioral. On the 320-entity evaluation subset per model, we sampled five answers per real entity and judged all of them with a strict factual-correctness rubric (claude-opus-4-8; 4,800 calls). Figure 3: Bielik-11B’s per-decile correctness rises monotonically from 0.06 (decile 0) to 0.68 (decile 9), mirroring the internal score’s gradient. Both correctness and the internal score share popularity as a plausible common cause, so we read the mirror as consistency between the two gradients; whether familiarity predicts correctness beyond pageviews and domain is an incremental-validity question we leave open (§10). Qwen3-14B is near floor everywhere (0.16 even at decile 9): it answers every Polish long-tail question with confident confabulation. PLLuM-12B is weak except at the very head (0.43 at decile 9). Gemma-4-12B is flat, but confounded: it is the only model that refuses, and the strict judge marks explicit refusals as factually correct $\sim 88\%$ of the time. Conditioning on *answering*, Gemma-4’s correct-of-non-refused rises

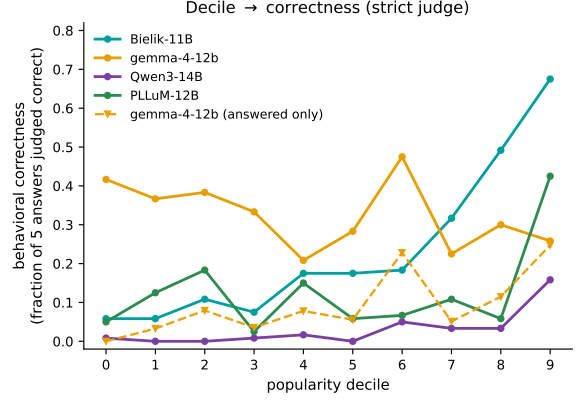


Figure 3: Behavioral mirror of the internal gradation: fraction of five sampled answers judged factually correct (strict LLM judge), per popularity decile. Bielik-11B rises monotonically 0.06 $\rightarrow$ 0.68 from decile 0 to 9; Qwen3-14B confabulates near-floor everywhere; Gemma-4-12B is flat because refusals are judged “correct”; conditioning on answering restores the gradient (0.00 $\rightarrow$ 0.25).

from 0.00 (decile 0) to 0.25 (decile 9), noisy but directionally consistent. A second judge outside the Claude family (gpt-5.2, byte-identical rubric) re-judged a stratified 20% of these verdicts (960 answers): pooled agreement is 89.8% at the answer level ($\kappa = 0.646$) and 91.7% at the entity level ($\kappa = 0.693$), and Bielik-11B’s decile-correctness monotone survives relabeling (Spearman 0.914 vs. 0.906 on the identical subset). The one low-agreement cell (Gemma-4, $\kappa = 0.469$) is precisely the refusal-scoring divergence above: the second judge scores refusals as correct far less often (positive rate 0.204 vs. Opus’s 0.371), corroborating the confound diagnosis. The mirror is not merely a popularity artifact: familiarity retains predictive value for behavioral error after controlling for log-pageviews and domain (likelihood-ratio $p = 0.013$ for Bielik-11B, $p < 0.001$ for Qwen3-14B), and a calibrated probe gate outperforms a popularity-only gate on Bielik (AURC 0.662 vs. 0.606).

6 RQ1: The Signal Is Robust to a Polish/English Stem Substitution

If the familiarity signal were a property of the question’s surface form (Polish morphology, tokenization quirks, template wording), it should degrade when the question language changes. We kept every entity string unchanged and switched only the question stem (*Kim jest ... ?* $\rightarrow$ *Who is ... ?*), then trained the probe in one language at its

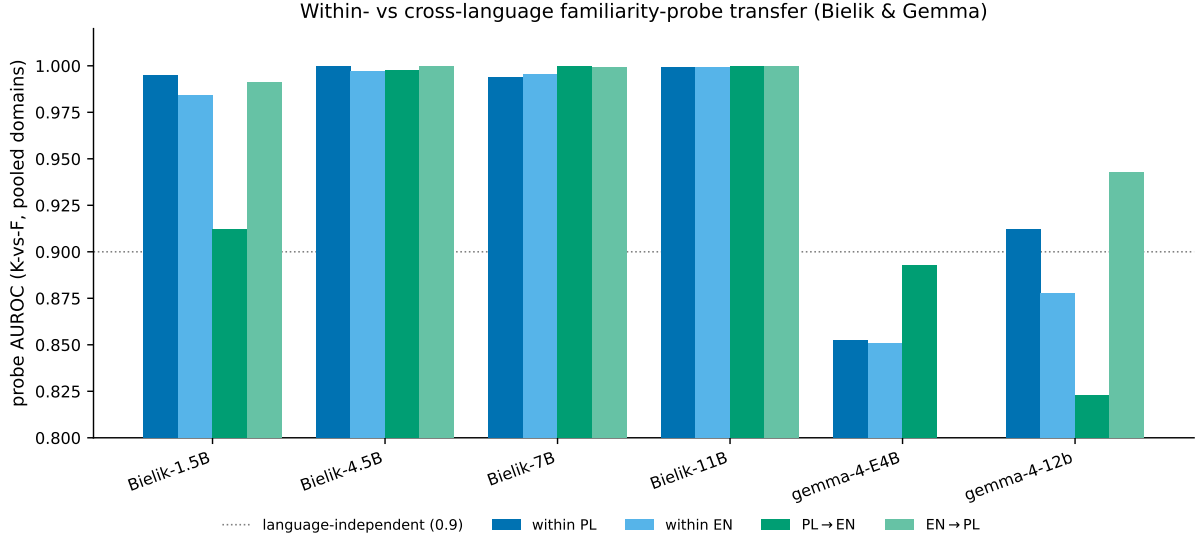


Figure 4: Cross-language probe transfer (v1 entities, six models, four domains). A probe trained on Polish questions and applied zero-shot to English questions about the *same* entities (and vice versa) retains 96–101% of within-language AUROC on known-vs-fabricated (per-model means) and 92–100% on the fabricated-string-free known-vs-obscure contrast.

CV-best layer and scored the other language zero-shot at the same layer, on identical entities (six models: four Bielik scales, two Gemma-4 scales; four domains; v1 conditions).

Figure 4: pooled over domains, cross-language AUROC retains 96–101% of within-language AUROC on known-vs-fabricated for every model (Bielik 0.961–1.005; Gemma-4 0.983–0.985), and 92–100% on the fabricated-string-free known-vs-obscure contrast. Every model is robust to the stem substitution in this paired setting, consistent with reports of language-agnostic latent processing in multilingual transformers (Wendler et al., 2024). Two design limits bound what this shows: the entity name, the largest semantic part of the prompt, is unchanged across conditions, and the cross-language evaluation reuses the entities the probe was trained on in the other language. To test transfer to unseen entities we re-ran the probe with entity-disjoint 5-fold splits (train one language on one entity set, test the other language on held-out entities, both directions). Transfer retains 98–100% of within-language AUROC on the same held-out split for the four Bielik scales, but only 74–93% for the two Gemma-4 scales: robustness to the stem swap generalizes to unseen entities in the Polish-adapted family and degrades for the multilingual model. A single template per language remains a limit (§10).

The design also argues against a specific al-

ternative: if Gemma-4’s weaker Polish signal reflected an English-centric model struggling with Polish *questions*, English stems should strengthen its within-language signal. They do not (mean probe delta EN–PL: -0.034 to -0.001 across contrasts and scales). What limits Gemma-4 here is its familiarity with the *entities*, not the question language. One caveat is inherent: the language switch also changes the template. A within-Polish neutral-template control in the predecessor study showed the signal is robust to a template swap at these scales, which bounds, but does not eliminate, this confound (§10).

7 RQ3: The Familiarity Direction Is Causally Potent

Correlational prelude. On Gemma-4-12B, the only model with a refusal behavior to predict, we first asked whether refusal is readable from the familiarity signal (v1 athletes, 42 entities per condition). Within the well-known condition it is: dispersion at the *independently chosen* known-vs-fabricated layer ranks refusals at AUROC 0.955 [0.878, 1.000] (zero-leakage probe 0.910; graded Spearman $\rho \approx -0.55$): the seven “known” entities Gemma-4 refuses are concentrated among those its activations assign lower familiarity. Within the unfamiliar conditions the mid-band score does not rank refusals (AUROC ~ 0.6 , CIs spanning 0.5), yet refusal remains strongly decodable from *late*

P0.1 (gemma-4-12b): the refusal decision thresholds the familiarity signal

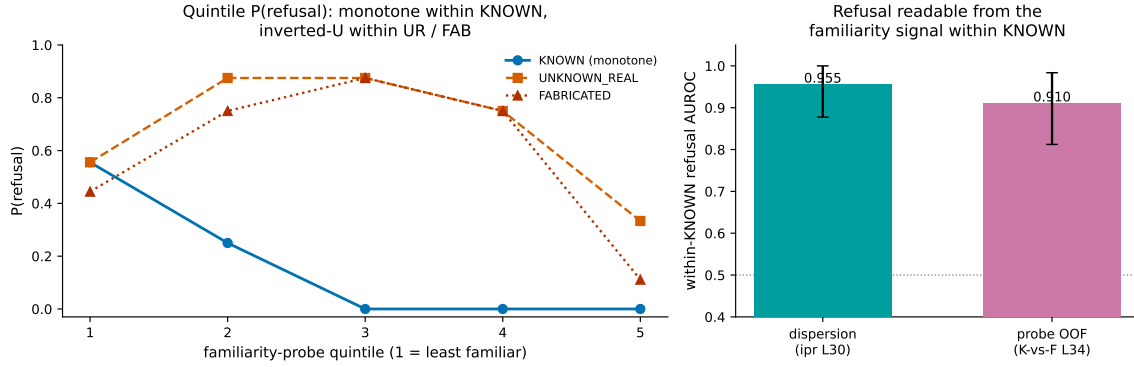


Figure 5: Correlational prelude (Gemma-4-12B, v1 athletes): within well-known entities, refusal is readable from the familiarity signal at the independently chosen familiarity layer (dispersion AUROC 0.955), and graded ($\rho \approx -0.55$): refusals are concentrated among the known entities assigned lower familiarity. Within unknown/fabricated conditions the mid-band familiarity score does not rank refusals, yet refusal remains strongly decodable from late prompt-point layers, motivating a separate late-layer refusal direction.

Causal steering (gemma-4-12b-it): the familiarity direction sets the refusal decision

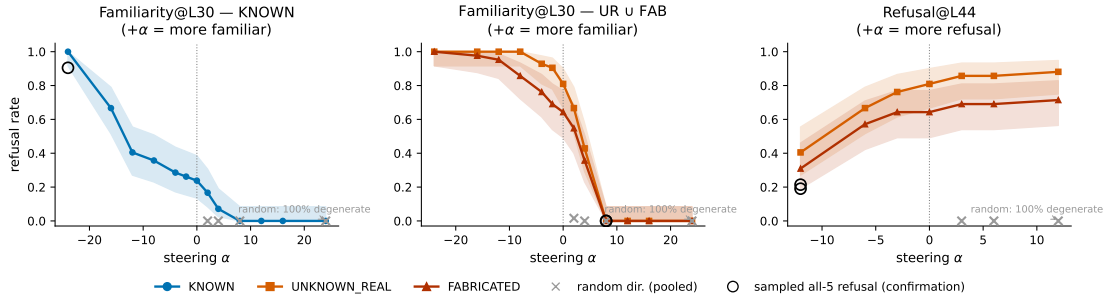


Figure 6: Causal dose-response (Gemma-4-12B, rank-one activation addition, single layer, greedy decoding). Steering along the familiarity direction (L30) moves refusal monotonically in both directions: to 1.00 on well-known entities at $\alpha = -24$ and to 0.00 on unknown/fabricated entities at $\alpha \geq +8$; no output triggered our four heuristic degeneration checks at any dose. Equal-norm random directions (pooled, three seeds) never produce a single Polish refusal and instead destroy output coherence. The late refusal direction (L44) is an independent, mostly suppressive handle.

prompt-point layers (out-of-fold 0.96–0.98; raw dispersion 0.89–0.94 at L42–47; Figure 5), suggesting a mid-depth familiarity representation and a late abstention readout downstream of it.

Intervention. Correlation alone cannot settle RQ3, so we intervene. We extracted two rank-one directions from prompt-point activations (difference of condition means, unit norm): a *familiarity* direction at layer 30 (mean(known) – mean(unknown-real $\cup$ fabricated)) and a *refusal* direction at layer 44 (mean(refused) – mean(non-refused) within the unfamiliar conditions). The two directions are geometrically distinct: cosine -0.14 , against an SD of 0.016 for random unit pairs in this residual space, a modest

negative alignment rather than a shared axis, so the familiarity direction is not a re-labeled refusal direction. During generation we add $\alpha \cdot \text{rms}(h) \cdot d$ to the residual stream at every position of the single target layer, where $\text{rms}(h) = \|h\|_2 / \sqrt{3840}$ keeps doses comparable across layers (cf. Turner et al., 2023; Rimsky et al., 2024; Arditì et al., 2024). We ran 5,040 greedy generations over a 13-point dose grid, with three equal-norm random directions per layer as controls and a four-way degeneration guard (empty, repetitive, non-Polish, token-salad outputs).

Results (Figure 6, Table 2). Pushing well-known entities toward “unfamiliar” ($\alpha < 0$ along familiarity) raises refusal monotonically

Condition	Refusal rate [Wilson 95% CI]		
	$\alpha=0$	fam. $\alpha=+8$	fam. $\alpha=-24$
Known	0.24 [.14,.39]	0.00 [.00,.08]	1.00 [.92,1.0]
Unknown-real	0.81 [.67,.90]	0.00 [.00,.08]	—
Fabricated	0.64 [.49,.77]	0.00 [.00,.08]	—
<i>Refusal direction, L44</i>			
		$\alpha=0$	$\alpha=-12$
Fabricated	0.64 [49,.77]	0.31 [19,.46]	
Unknown-real	0.81 [67,.90]	0.40 [27,.56]	

Table 2: Steering key cells (Gemma-4-12B, athletes, $n=42$ per condition). Familiarity steering at L30 saturates refusal in both directions; the late refusal direction at L44 suppresses refusal without touching familiarity. Random equal-norm controls (3 seeds, pooled): refusal 0.00 at every amplitude, but with 100% of outputs flagged by the degeneration heuristics; no output from the trained directions is flagged.

from 0.238 to 1.000 at $\alpha = -24$ (dose-curve Spearman $\rho = -0.99$); pushing unknown and fabricated entities toward “familiar” suppresses refusal from 0.726 to 0.000 at $\alpha \geq +8$ ($\rho \leq -0.97$). The model then fluently invents biographies: a confabulation proxy (coherent non-refusal answers about unknown entities) rises from 0.27 to 1.00. No steered output triggers any of the four degeneration heuristics; the heuristics screen for gross corruption (empty, repetitive, non-Polish, token-salad outputs) and do not certify preserved semantics or factuality beyond that. Random equal-norm directions never produce a single Polish refusal at any amplitude and instead destroy the output (degeneration 0.39–1.00; at the amplitudes where familiarity steering saturates refusal, random controls are 100% degenerate), so the effect is not a generic norm artifact. Random directions are a coarse specificity control, however: they show that arbitrary directions are destructive at these amplitudes, not that the familiarity direction is the unique semantic direction with this effect. Five-sample confirmations at temperature 0.7 reproduce the endpoints (known at $\alpha = -24$: mean refusal 0.976, all 42 entities refuse at least once; unfamiliar at $\alpha = +8$: mean refusal 0.019–0.024), and a 60-generation external audit of the refusal marker (claude-haiku-4-5) agrees with the marker metric at 0.95. These results converge with Ferrando et al. (2025), who causally modulated refusal via entity-recognition SAE directions in a Gemma 2 chat model; we add the

dose–response characterization, equal-norm random controls with degeneration accounting, sampled confirmations, and the late-layer refusal direction below.

The late refusal direction, and a reframing.

The L44 refusal direction is causally potent but asymmetric: pushing *away* from refusal works ($0.726 \rightarrow 0.357$ at $\alpha = -12$; confirmed at five samples), pushing *toward* refusal adds only +0.05 to +0.07 (baseline 0.73–0.81 is near ceiling), and on well-known entities it does nothing. The strict hypothesis motivating it, that the mid-band familiarity direction is causally inert where it fails to *rank* refusals, is falsified: familiarity steering saturates refusal in both directions within the unfamiliar conditions too. Correlational dissociation does not imply causal inertness. The familiarity direction is causally potent for refusal in all three conditions; the late refusal representation adds an independently steerable, mostly suppressive handle. We note the scope of this claim: the intervention is applied at every generation position, not only at the prompt point, and both directions are estimated on the same 42 entities per condition they are evaluated on. A held-out-entity check (correlational, saved activations only) shows the direction is not an artifact of the specific entities: built on a training subset, it separates known from unfamiliar held-out entities at AUROC 0.800 versus 0.802 in-sample, and train-fold directions have cosine 0.999 with the full-set direction. What is demonstrated is that a one-dimensional direction derived from familiarity contrasts strongly steers Gemma-4-12B’s refusal behavior; showing that the model’s *natural* prompt-time familiarity computation feeds the refusal decision would additionally require prompt-point-only interventions on held-out entities, or mediation-style evidence (§10).

8 RQ4: Gating Generation with a One-Pass Signal

Setup. The applied question is whether the one-pass signal can *gate*: decide, before generating, whether to answer from memory, retrieve, or abstain. On the 320-entity evaluation subset per model (six real entities per decile plus 20 fabricated, per domain; stratified, fixed seed) we compare our probe (Platt-calibrated logistic probe on prompt-point residuals, trained only on *non-eval* v2 rows; it never sees an evalu-

Method (cost)	Bielik-11B	Gemma-4-12B	Qwen3-14B	PLLuM-12B
<i>(a) Real-vs-fabricated entity discrimination</i>				
Our probe (1 pass)	0.934 [.91,.96]	0.871 [.83,.91]	0.859 [.81,.91]	0.903 [.87,.94]
Our dispersion (1 pass)	0.746 [.69,.80]	0.657 [.59,.73]	0.666 [.60,.73]	0.707 [.64,.77]
First-token entropy (1 pass)	0.652 [.59,.72]	0.560 [.49,.62]	0.407 [.34,.48]	0.627 [.56,.70]
Semantic entropy (5 gen + API)	0.750 [.69,.80]	0.477 [.40,.54]	0.480 [.41,.55]	0.585 [.52,.65]
D ² HScore (1 gen) ^a	0.629 [.55,.69]	0.438 [.36,.51]	0.506 [.44,.58]	0.489 [.42,.56]
EigenTrack (1 gen, sup.) ^b	0.621 [.55,.69]	0.648 [.58,.71]	0.535 [.46,.61]	0.443 [.36,.52]
MIND (1 gen, sup.) ^c	0.653 [.58,.72]	0.677 [.61,.74]	0.717 [.65,.78]	0.601 [.53,.67]
<i>(b) Behavioral: majority-incorrect vs. correct (strict judge, real entities only)</i>				
Our probe (1 pass)	0.764 [.70,.83]	0.409 [.34,.48]	0.914 [.80,.99]	0.722 [.62,.81]
Our dispersion (1 pass)	0.675 [.58,.77]	0.262 [.20,.34]	0.866 [.64,1.0]	0.656 [.52,.78]
First-token entropy (1 pass)	0.661 [.57,.75]	0.355 [.29,.43]	0.441 [.24,.67]	0.629 [.51,.74]
Semantic entropy (5 gen + API)	0.805 [.74,.86]	0.761 [.71,.81]	0.806 [.64,.95]	0.782 [.67,.89]
D ² HScore (1 gen) ^a	0.696 [.62,.77]	0.768 [.70,.83]	0.409 [.21,.61]	0.229 [.13,.33]
EigenTrack (1 gen, sup.) ^b	0.684 [.60,.77]	0.748 [.68,.81]	0.503 [.26,.73]	0.490 [.36,.63]
MIND (1 gen, sup.) ^c	0.528 [.44,.62]	0.778 [.72,.84]	0.782 [.52,.98]	0.858 [.79,.92]

Table 3: AUROC (1000-bootstrap 95% CIs) of one-pass signals vs. literature baselines on the v2 evaluation subset ($n=320$ per model; contrast (a): 80 fabricated positives; contrast (b): majority-of-5-answers incorrect vs. correct under the strict judge, real entities only). Block (b) is heavily imbalanced for Qwen3 (7 majority-correct of 240) and PLLuM (25 of 240); paired bootstrap: in block (a) the probe’s margin over the best baseline excludes zero on every model, in block (b) no best-vs-runner-up difference does, so bold in (b) is numerical, not statistical. Baseline fidelity tiers: ^aD²HScore is training-free and runnable exactly as specified, but its drift component is anti-predictive in our one-sentence-answer regime; ^bEigenTrack instantiates 16 of ~ 22 unspecified spectral features with a window shortened for short answers, trained with 5-fold CV *on eval-pool entities*; ^cMIND’s detector (feature + architecture) is trained supervised with CV; its defining unsupervised Wikipedia-continuation data pipeline is not reproduced, so these numbers reflect our adapted variant, not the published method, and the direction of the net bias is unknown. The supervised baselines see 4/5 of the eval pool in training, while our probe never sees any eval entity (trained on disjoint non-eval v2 rows); this supervision asymmetry favors the baselines, but the fidelity gaps above have unknown sign.

ated entity) and dispersion against first-token entropy, discrete semantic entropy over five samples (Kuhn et al., 2023; Farquhar et al., 2024), and three post-generation internal-state detectors: D²HScore (Ding et al., 2025), EigenTrack (Ettori et al., 2025), and MIND (Su et al., 2024), reimplemented from the papers with all adaptations documented in the caption of Table 3 and the released code.

Real-vs-fabricated discrimination. On fabricated-vs-real (Table 3a) the calibrated probe outperforms our adapted implementations of every baseline on every model (0.859–0.934), beating the best generation-time baseline by 0.14–0.30 AUROC per model (paired bootstrap over entities: every per-model margin’s 95% CI excludes zero), despite the supervised baselines training on eval-pool entities while the probe never does. On this target, the prompt point is easier to read than the generated answer’s internal states. Purely lexical classifiers on the entity string top out at AUROC 0.786 (character n-grams; single length and token-count features

stay ≤ 0.61 under all four family tokenizers), so the probe’s 0.859–0.934 sits 0.07–0.15 above the lexical-naturalness ceiling and is not explained by surface name form. We stress that the target is operational: it certifies discrimination of our real/fabricated construction, not possession of answer-relevant knowledge; a model can register that an entity exists while knowing no usable facts about it (§10).

Behavioral prediction and the Gemma-4 inversion. On predicting strict-judge incorrectness (Table 3b) semantic entropy is the most consistent (0.76–0.81 on all four models): it directly measures answer instability. The probe transfers well where behavior follows knowledge (Qwen3 0.914, Bielik 0.764), though no per-cell best-vs-runner-up difference in this contrast is individually significant under a paired bootstrap (e.g., Qwen3 probe-vs-MIND CI $[-0.12, +0.39]$), so we read block (b) as pattern-level rather than cell-level evidence. On Gemma-4 it appears to *invert* (0.409), but this is a label-semantics artifact, not a property of the signal: the strict judge scores explicit

Selective answering (risk-coverage): pre-generation probe gates before any generation cost

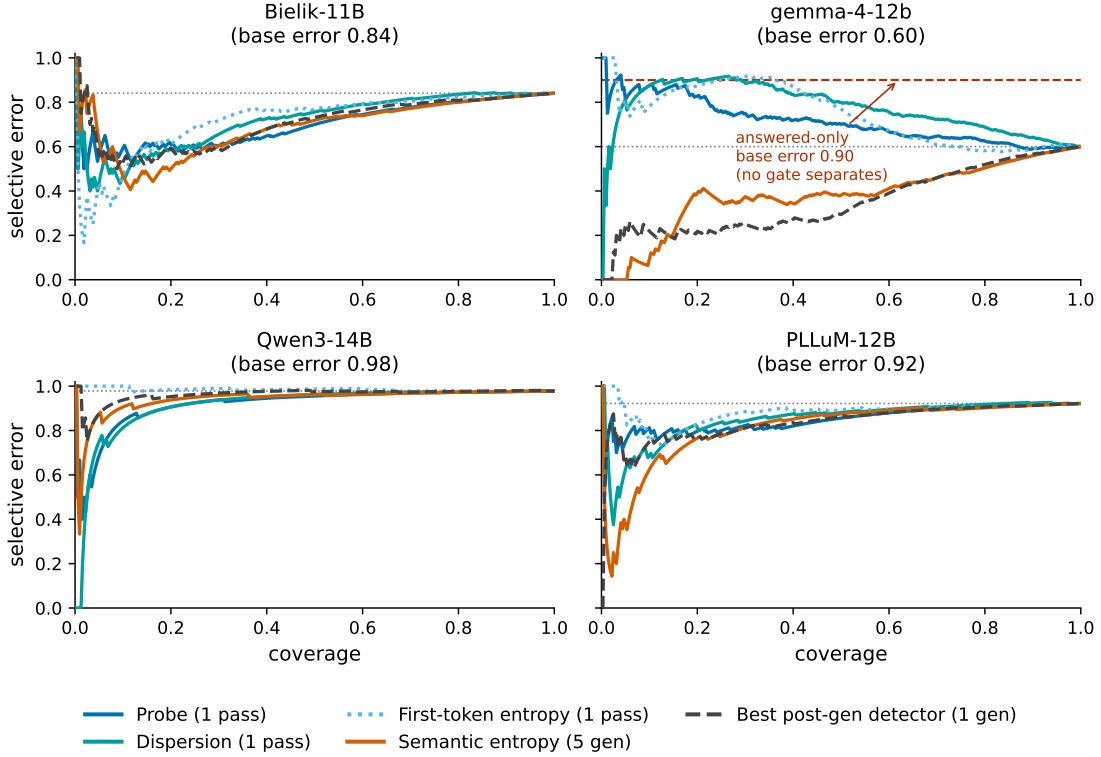


Figure 7: Risk-coverage curves for selective answering on the Polish long tail (target: entity answered incorrectly under the strict judge; for Gemma-4, refusals judged “correct” lower its apparent base error, §10). Base error without gating is severe (0.60–0.98 across models). The calibrated one-pass probe is the best or effectively tied *pre-generation* gate on three of four models; the post-generation detectors pay off only on Gemma-4, i.e. only when the model’s own generation behavior (refusal) already separates the classes.

refusals as correct, and Gemma-4’s refusals fall mostly on unfamiliar entities. Separating refusal from correctness with the published marker list (which reconciles to Table 5’s 533/1200) shows the inversion is a labeling artifact, not a signal property: among *answered* items Gemma-4’s base error is 0.90, and only 13 of 149 answered entities are majority-correct, too few to estimate a behavioral AUROC in either direction. The dissociation that matters here is therefore carried by the refusal behavior itself (Table 5), not by an inverted AUROC: the familiarity signal tracks the entity’s status under our operational measure; behavioral error is additionally shaped by the model’s policy. More generally, folding refusals into “correct” conflates abstention with factual accuracy, and every block (b) number, and the risk-coverage analysis below, inherits this target definition for the one abstaining model; a clean analysis would separate answered-vs-abstained from correctness-given-answering, which we flag wher-

ever Gemma-4 is affected and leave as the principal evaluation revision.

Selective answering. Framed as risk-coverage (El-Yaniv and Wiener, 2010; Geifman and El-Yaniv, 2017) (Figure 7, Table 4), base error rates on this long-tail benchmark are severe: 0.84 (Bielik-11B), 0.60 (Gemma-4), 0.98 (Qwen3), 0.92 (PLLuM); Gemma-4’s 0.60 counts its refusals as correct, and among *answered* items its base error is 0.90 (§10). The gates also differ in inference category (Figure 8; we report pass and generation counts rather than measured latency or FLOPs): semantic entropy achieves the best macro AURC (0.708) but requires five sampled generations plus an external clustering call and can only act *after* answering. Every post-generation detector beats the probe’s macro AURC (0.708–0.773 vs. 0.797). The probe’s claim is a within-class one: it is the best or effectively tied gate in the pre-generation class on three of four models (the PLLuM margin over dispersion sits at the fifth

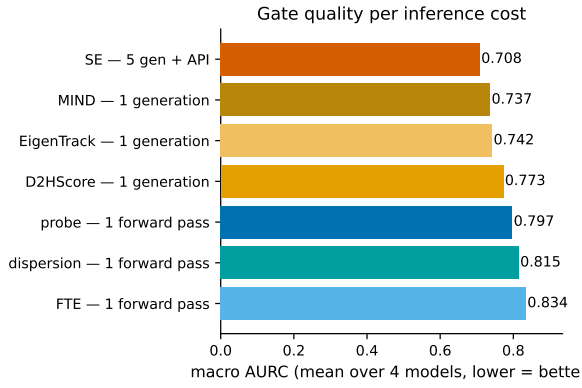


Figure 8: Gate quality by inference category (we report pass/generation counts, not measured cost). Semantic entropy attains the best macro AURC (0.708) but requires five sampled generations plus an external clustering call and cannot act before generation; the calibrated probe (macro 0.797) is the strongest gate in the one-forward-pass, pre-generation class, where routing decisions (answer from memory vs. retrieve) must be made.

decimal; on Qwen3, dispersion is better by 0.007), and pre-generation is the only class that can route before generating. The post-generation detectors beat the probe only on Gemma-4, but this advantage is largely a labeling artifact: once refusals are treated as abstentions and risk is measured among answered items, no gate separates on Gemma-4 (AURC 0.83–0.93, against a 0.90 answered base error). Calibration: against the behavioral target the probe’s raw ECE is 0.41–0.64, a target mismatch (it is calibrated to the real-vs-fabricated target, not answer incorrectness); a one-dimensional out-of-fold Platt recalibration (Platt, 1999; Guo et al., 2017) onto the behavioral target yields ECE 0.039 / 0.063 / 0.000 / 0.049 on Bielik/Gemma-4/Qwen3/PLLuM. At these base rates a constant predictor attains ECE 0.000 by construction, so low recalibrated ECE certifies that calibration is achievable, not that the score discriminates; ranking quality is carried by the AURC analysis above.

Who acts on the signal? Table 5: only Gemma-4 refuses at all, and selectively by domain type: 52–64% per-answer refusal on real people vs. 3% on cities, a biography-caution policy layered on top of the knowledge signal. Bielik, Qwen3, and PLLuM produce zero refusals in 1,600 answers each: whatever familiarity information their activations carry, their generation policies never convert it into abstention.

Gate (cost)	B-11B	G-12B	Q-14B	P-12B	macro
SE (5g+API)	0.697	0.381	0.944	0.811	0.708
MIND (1g, sup.)	0.785	0.353	0.967	0.841	0.737
EigenTrack (1g, sup.)	0.712	0.383	0.964	0.909	0.742
D ² HScore (1g)	0.728	0.405	0.986	0.972	0.773
Probe (1p, pre-gen)	0.701	0.712	0.923	0.852	0.797
Dispersion (1p)	0.719	0.771	0.916	0.852	0.815
FT entropy (1p)	0.722	0.745	0.985	0.886	0.834
Base error (no gate)	0.841	0.600	0.978	0.922	—

Table 4: AURC (mean selective error over all coverage prefixes; lower is better; 1g = one generation, 1p = one forward pass, 5g = five sampled generations). **Bold** marks the best gate *within the pre-generation class* per model, the only class that can route before generating; probe and dispersion tie on PLLuM at the reported precision (both bolded). All four post-generation detectors have better *macro* AURC (0.708–0.773) than the probe (0.797); their advantage is concentrated on Gemma-4, the sole abstaining model, under a strict-judge target that scores refusals as correct (its 0.60 base error). With refusals removed and risk measured among answered items, Gemma-4’s base error is 0.90 and no gate separates (AURC 0.83–0.93), so that advantage largely disappears (§10).

9 Discussion

The four results compose into one claim: **prompt-point activations carry a graded entity-familiarity readout that is robust to a Polish/English stem substitution, is more strongly associated with Polish adaptation than with scale in this sample, and can strongly steer the one native abstention policy we observe.** Gradation (§5) shows the quantity co-varies with Polish adaptation rather than parameter count, observationally. Language transfer (§6) argues against a large question-stem-language effect in the paired setting, consistent with entity-level recognition features (Ferrando et al., 2025); entity-disjoint transfer remains to be tested. Steering (§7) shows the familiarity direction is causally potent for refusal, though establishing that the model’s natural prompt-time computation flows through it requires mediation-style evidence. The gating experiments (§8) show the value and the boundary: the signal separates real from fabricated entities better than our adapted post-generation baselines and can act before a single token is generated, but predicting *behavioral* error additionally requires the model’s policy, which varies from Gemma-4’s domain-selective caution to the absence of refusal behavior in the other three families. What distin-

Model	Answers refused		Per-answer refusal	
	real	fabr.	people	cities
Bielik-11B	0/1200	0/400	0.00	0.00
Gemma-4-12B	533/1200	246/400	0.52–0.64	0.03
Qwen3-14B	0/1200	0/400	0.00	0.00
PLLuM-12B	0/1200	0/400	0.00	0.00

Table 5: Refusal patterns on the v2 evaluation subset (marker-based). Only Gemma-4 abstains, and selectively by *domain type*: 52–64% per-answer refusal on real people (athletes 0.58, writers 0.64, musicians 0.52) vs. 3% on cities, a biography-caution policy rather than a pure knowledge test. Polish continual pretraining (PLLuM) did not install any abstention policy: zero marker hits in 1,600 answers.

guishes a family that consumes the signal from one that leaves it inert, for instance whether the familiarity direction is coupled into the model’s output pathway, is an open mechanistic question we are pursuing in ongoing work.

The dissociation has a deployment corollary for Polish and other medium-resourced languages. Polish-adapted models carry more graded familiarity information but never abstain; Gemma-4 abstains but its readout is less graded. A one-forward-pass calibrated probe supplies the missing piece externally: a familiarity-aware gate that routes long-tail queries to retrieval (cf. Asai et al., 2024) before a single token is generated. The representational prerequisite for calibrated abstention is thus present in every family we measured; what is missing in three of them is a policy that consumes it. Whether such a policy can be *installed* in a never-refusing model, by activation-level intervention or by the refusal-aware tuning studied for English models (Zhang et al., 2024; Cheng et al., 2024), is the natural next question; we leave a systematic comparison to future work.

10 Limitations

Familiarity is an operational construct. Our primary discrimination target is real-vs-fabricated status, not validated knowledge: a model can register that an entity exists while holding no answer-relevant facts, and the probe’s positive training class (top popularity deciles) couples its score to the popularity axis by construction. The fabricated names are token-length matched, but we did not control character n-grams, name-component frequency, or morphological likelihood, so a lexical-naturalness shortcut cannot be fully excluded;

non-neural lexical baselines and real-entity-only controls (e.g., post-cutoff entities) are the appropriate follow-up. Lexical baselines bound the shortcut concern: character-n-gram models reach AUROC 0.786, so the probe’s 0.859–0.934 is above but not fully independent of surface form. Familiarity also retains predictive value for behavioral error after controlling for log-pageviews and domain (likelihood-ratio $p = 0.013/p < 0.001$ for Bielik-11B/Qwen3-14B), so the behavioral mirror is not purely a popularity common-cause, though popularity remains a strong baseline.

Single steering model and domain. The steering result is established on one model (Gemma-4-12B, the only one with a refusal behavior to steer) and one domain (athletes, $n=42$ per condition; Wilson CIs are wide); whether familiarity steering can *install* abstention in the never-refusing families is untested. Both directions are estimated from the entities they are tested on, so effect sizes may be optimistic; the intervention acts at every generation position, so it demonstrates the direction’s potency rather than isolating the natural prompt-point pathway; and the familiarity direction contrasts known entities with obscure-real and fabricated ones pooled, so it may encode correlated features (popularity, real/fabricated status, lexical form) alongside familiarity. High-dose specificity rests on the degeneration contrast plus low-dose non-degenerate controls, and the degeneration heuristics do not certify preserved semantics.

Judge circularity and the refusal label. Correctness labels come from a single model family (claude-opus-4-8). A second judge (gpt-5.2, identical rubric) covers a stratified 20% of the v2 verdicts at $\kappa = 0.65$ –0.69 (§5), substantial but not perfect; behavioral numbers inherit the residual uncertainty, and no human audit was performed. The strict judge scores explicit refusals as correct $\sim 88\%$ of the time, conflating abstention with factual accuracy for Gemma-4; this affects the block (b) AUROCs, base error rates, risk-coverage/AURC, and behavioral calibration wherever Gemma-4 appears. We report a marker-based separation of answered-vs-abstained from correctness-given-answering (§8): it reconciles to Table 5 and shows Gemma-4’s answered base error is 0.90, not the refusal-inflated 0.60, and that the post-generation gates’ Gemma-4 advantage is largely a labeling artifact. An LLM three-way re-judge (refuse/correct/incorrect; claude-opus-4-8; all 4,800 real-entity answers) confirms this: it

agrees with the marker list at 0.99 per answer (Gemma-4 542 refusals vs. the marker’s 533) and reproduces the answered-only picture (Gemma-4 base error 0.88, 14 of 131 answered entities majority-correct).¹ A human audit remains future work.

Baseline fidelity. Our MIND variant omits its unsupervised data pipeline and is trained supervised with CV (likely flattering it); EigenTrack’s unspecified feature set is instantiated as 16 features with a shortened window; D²HScore runs as specified but its drift term is anti-predictive for one-sentence answers. All adaptations are documented and the supervision asymmetry favors the baselines.

Subsample sizes. Gating and behavioral results use 320 entities per model; per-decile correctness estimates average 24 entities; the behavioral contrast is heavily imbalanced for Qwen3 (7 majority-correct of 240) and PLLuM (25/240).

Popularity proxy. Pageviews proxy pretraining exposure imperfectly: the 12-month window is contemporary while the models’ training cutoffs differ and predate it, and pageviews correlate with article length, link density, and recency. The bottom decile contains zero-pageview stubs with no internal gradient; sitelink counts give materially identical results.

Family-level inference. Only four model families are compared, and checkpoints within a family are correlated replications, so the exposure-vs-scale conclusion is sample-limited (§5). We add controlled before/after evidence for both PLLuM architectures: the Mistral-NeMo-12B and Llama-3.1-8B bases grade at $\rho = 0.102$ and 0.193 versus their descendants’ 0.277 and 0.359 , isolating Polish continual pretraining from architecture and scale.

Refusal metric. Refusal is detected by a Polish marker list validated by two independent audits (agreement 0.95–1.00), but steering-induced

phrasing outside the list would be missed; the metric is untested on other abstention styles.

Language-transfer scope. The EN/PL switch changes both language and template, and only one template per language is tested; a within-Polish neutral-template control bounds the template effect but was not repeated for every v2 model. Entity strings remain Polish in both conditions by design, so EN prompts are code-mixed rather than fully English. We do test entity-disjoint transfer (§6); it holds for Bielik but weakens for Gemma-4. Multiple paraphrases, translated entity forms where aliases exist, and additional languages are still needed before any language-independence claim.

Ethics and Availability Statement

We use only public information about public entities, plus invented names screened against real-person collisions. Steering shows abstention can be *suppressed* as easily as induced, so deployed abstention policies are not robust to activation-level manipulation; familiarity gating abstains more on long-tail entities, a coverage-disparity risk. Code, the CSV dataset, and provenance artifacts, including pinned model checkpoints and judge identifiers, are available online.² The experiments were orchestrated with AI coding assistance. All results were produced and verified by the author.

References

- Riasad Alvi, Nurul Labib Sayeedi, and Md. Faiyaz Abdullah Sayeedi. 2026. [MultiHaluDet: Multilingual hallucination detection via LLM hidden state probing](#). In *Proceedings of the 1st Workshop on Multilinguality in the Era of Large Language Models (MeLLM 2026)*. Association for Computational Linguistics.
- Andy Arditi, Oscar Obeso, Aaqib Syed, Daniel Paleka, Nina Panickssery, Wes Gurnee, and Neel Nanda. 2024. [Refusal in language models is mediated by a single direction](#). In *Advances in Neural Information Processing Systems 37 (NeurIPS 2024)*.
- Akari Asai, Zeqiu Wu, Yizhong Wang, Avirup Sil, and Hannaneh Hajishirzi. 2024. [Self-RAG: Learning to retrieve, generate, and critique through self-reflection](#). In *The Twelfth International Conference on Learning Representations (ICLR)*.

¹Binary and three-way judges agree on 86% of the 4,800 answers (cf. the 89.8% cross-family agreement, §5). Disagreements are of two kinds. The binary judge splits the 543 refusals inconsistently (327 scored correct, 216 incorrect), where the three-way judge labels all `refuse`; this is the source of the correction. The other 142 are borderline factual answers on long-tail entities (133 flip incorrect→correct, 9 the reverse), a small non-directional shift: the same entity often draws opposite flips on different samples, so per-entity majority labels and Bielik’s decile monotone are unchanged. The residual correct/incorrect variation is thus judge noise of the same order as the cross-family disagreement, not a systematic bias.

²<https://github.com/agentGreg/bielik-entity-familiarity>

- Amos Azaria and Tom Mitchell. 2023. [The internal state of an LLM knows when it’s lying](#). In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 967–976. Association for Computational Linguistics.
- Jakub Binkowski, Denis Janiak, Albert Sawczyn, Bogdan Gabrys, and Tomasz Kajdanowicz. 2025. [Hallucination detection in LLMs using spectral features of attention maps](#). In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP)*.
- Grzegorz Brzezinka. 2026. [Does Bielik know what it doesn’t know? activation dispersion separates entity familiarity from factual reliability across model scale](#). ArXiv:2607.07670.
- Collin Burns, Haotian Ye, Dan Klein, and Jacob Steinhardt. 2023. [Discovering latent knowledge in language models without supervision](#). In *The Eleventh International Conference on Learning Representations (ICLR)*.
- Chi Seng Cheang, Hou Pong Chan, Wenxuan Zhang, and Yang Deng. 2026. [Do LLMs really know what they don’t know? internal states mainly reflect knowledge recall rather than truthfulness](#). Findings of ACL 2026.
- Chao Chen, Kai Liu, Ze Chen, Yi Gu, Yue Wu, Mingyuan Tao, Zhihang Fu, and Jieping Ye. 2024. [INSIDE: LLM’s internal states retain the power of hallucination detection](#). In *International Conference on Learning Representations (ICLR)*.
- Qinyuan Cheng, Tianxiang Sun, Xiangyang Liu, Wenwei Zhang, Zhangyue Yin, Shimin Li, Linyang Li, Zhengfu He, Kai Chen, and Xipeng Qiu. 2024. [Can AI assistants know what they don’t know?](#) In *Proceedings of the 41st International Conference on Machine Learning (ICML)*.
- Yue Ding, Xiaofang Zhu, Tianze Xia, Junfei Wu, Xinlong Chen, Qiang Liu, and Liang Wang. 2025. [D²HScore: Reasoning-aware hallucination detection via semantic breadth and depth analysis in LLMs](#). Under review.
- Xuefeng Du, Chaowei Xiao, and Yixuan Li. 2024. [HaloScope: Harnessing unlabeled LLM generations for hallucination detection](#). In *Advances in Neural Information Processing Systems (NeurIPS)*.
- Ran El-Yaniv and Yair Wiener. 2010. On the foundations of noise-free selective classification. *Journal of Machine Learning Research*, 11:1605–1641.
- Davide Etti, Nastaran Darabi, Sina Tayebati, Ranganath Krishnan, Mahesh Subedar, Omesh Tickoo, and Amit Ranjan Trivedi. 2025. [EigenTrack: Spectral activation feature tracking for hallucination and out-of-distribution detection in LLMs and VLMs](#). Submitted to ICASSP 2026.
- Sebastian Farquhar, Jannik Kossen, Lorenz Kuhn, and Yarin Gal. 2024. [Detecting hallucinations in large language models using semantic entropy](#). *Nature*, 630(8017):625–630.
- Javier Ferrando, Oscar Obeso, Senthooan Rajamanoharan, and Neel Nanda. 2025. [Do I know this entity? Knowledge awareness and hallucinations in language models](#). In *International Conference on Learning Representations (ICLR)*.
- Yonatan Geifman and Ran El-Yaniv. 2017. [Selective classification for deep neural networks](#). In *Advances in Neural Information Processing Systems 30 (NIPS 2017)*.
- Gemma Team, Google DeepMind. 2026. Gemma 4 model card. https://ai.google.dev/gemma/docs/core/model_card_4. Models google/gemma-4-E4B-it and google/gemma-4-12b-it; no technical report was available at the time of writing.
- Daniela Gottesman and Mor Geva. 2024. [Estimating knowledge in large language models without generating a single token](#). In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 3994–4019. Association for Computational Linguistics.
- Aaron Grattafiori et al. 2024. [The llama 3 herd of models](#).
- Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. 2017. [On calibration of modern neural networks](#). In *Proceedings of the 34th International Conference on Machine Learning (ICML)*.
- Saurav Kadavath, Tom Conerly, Amanda Askell, Tom Henighan, Dawn Drain, Ethan Perez, Nicholas Schiefer, Zac Hatfield-Dodds, Nova DasSarma, Eli Tran-Johnson, Scott Johnston, Sheer El-Showk, Andy Jones, Nelson Elhage, Tristan Hume, Anna Chen, Yuntao Bai, Sam Bowman, Stanislaw Fort, Deep Ganguli, Danny Hernandez, Josh Jacobson, Jackson Kernion, Shauna Kravec, Liane Lovitt, Kamal Ndousse, Catherine Olsson, Sam Ringer, Dario Amodei, Tom Brown, Jack Clark, Nicholas Joseph, Ben Mann, Sam McCandlish, Chris Olah, and Jared Kaplan. 2022. [Language models \(mostly\) know what they know](#).
- Nikhil Kandpal, Haikang Deng, Adam Roberts, Eric Wallace, and Colin Raffel. 2023. [Large language models struggle to learn long-tail knowledge](#). In *Proceedings of the 40th International Conference on Machine Learning (ICML)*, volume 202 of *PMLR*, pages 15696–15707.
- Remigiusz Kinas, Paweł Kiszcak, Sergio P. Perez, Krzysztof Ociepa, Łukasz Flis, Krzysztof Wróbel, and Adrian Gwoździej. 2026. [Bielik-minitron-7b: Compressing large language models via structured pruning and knowledge distillation for the polish language](#). ArXiv:2603.11881.

- Jan Kocoń, Maciej Piasecki, Arkadiusz Janz, et al. 2025. [PLLuM: A family of polish large language models](#).
- Jannik Kossen, Jiatong Han, Muhammed Razzak, Lisa Schut, Shreshth Malik, and Yarin Gal. 2024. [Semantic entropy probes: Robust and cheap hallucination detection in LLMs](#). ICML 2024 Workshop on Foundation Models in the Wild.
- Lorenz Kuhn, Yarin Gal, and Sebastian Farquhar. 2023. [Semantic uncertainty: Linguistic invariances for uncertainty estimation in natural language generation](#). In *International Conference on Learning Representations (ICLR)*.
- Bruce W. Lee, Inkit Padhi, Karthikeyan Natesan Ramamurthy, Erik Miehl, Pierre Dognin, Manish Nagireddy, and Amit Dhurandhar. 2025. [Programming refusal with conditional activation steering](#). In *International Conference on Learning Representations (ICLR)*. Spotlight presentation.
- Kenneth Li, Oam Patel, Fernanda Viégas, Hanspeter Pfister, and Martin Wattenberg. 2023. [Inference-time intervention: Eliciting truthful answers from a language model](#). In *Advances in Neural Information Processing Systems 36 (NeurIPS)*.
- Stephanie Lin, Jacob Hilton, and Owain Evans. 2022. [Teaching models to express their uncertainty in words](#). *Transactions on Machine Learning Research*.
- Alex Mallen, Akari Asai, Victor Zhong, Rajarshi Das, Daniel Khashabi, and Hannaneh Hajishirzi. 2023. [When not to trust language models: Investigating effectiveness of parametric and non-parametric memories](#). In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 9802–9822. Association for Computational Linguistics.
- Potsawee Manakul, Adian Liusie, and Mark J. F. Gales. 2023. [SelfCheckGPT: Zero-resource black-box hallucination detection for generative large language models](#). In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. Association for Computational Linguistics.
- Samuel Marks and Max Tegmark. 2024. [The geometry of truth: Emergent linear structure in large language model representations of true/false datasets](#). In *First Conference on Language Modeling (COLM)*.
- Jack Merullo, Noah A. Smith, Sarah Wiegrefe, and Yanai Elazar. 2025. [On linear representations and pretraining data frequency in language models](#). In *International Conference on Learning Representations (ICLR)*.
- Mistral AI. 2024. Mistral NeMo. <https://mistral.ai/news/mistral-nemo/>. 12B open model developed with NVIDIA; base model of PLLuM-12B.
- Iván Vicente Moreno Cencerrado, Arnau Padrés Masdemont, Anton Gonzalvez Hawthorne, David Demetri Africa, and Lorenzo Pacchiardi. 2025. [No answer needed: Predicting LLM answer accuracy from question-only linear probes](#). Accepted (poster), Principled Design for Trustworthy AI Workshop, ICLR 2026.
- Jamshid Mozafari, Bhawna Piryani, and Adam Jawtowt. 2026. [Pretraining exposure explains popularity judgments in large language models](#). In *Proceedings of the 49th International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR)*.
- Oscar Obeso, Andy Arditi, Javier Ferrando, Joshua Freeman, Cameron Holmes, and Neel Nanda. 2025. [Real-time detection of hallucinated entities in long-form generation](#).
- Krzysztof Ociepa, Łukasz Flis, Remigiusz Kinas, Krzysztof Wróbel, and Adrian Gwoździej. 2025a. [Bielik v3 small: Technical report](#).
- Krzysztof Ociepa, Łukasz Flis, Krzysztof Wróbel, Adrian Gwoździej, and Remigiusz Kinas. 2025b. [Bielik 11b v2 technical report](#).
- Hadas Orgad, Michael Toker, Zorik Gekhman, Roi Reichart, Idan Szpektor, Hadas Kotek, and Yonatan Belinkov. 2025. [LLMs know more than they show: On the intrinsic representation of LLM hallucinations](#). In *International Conference on Learning Representations (ICLR)*.
- Seongheon Park, Xuefeng Du, Min-Hsuan Yeh, Haobo Wang, and Yixuan Li. 2025. [Steer LLM latents for hallucination detection](#). International Conference on Machine Learning (ICML) 2025.
- John C. Platt. 1999. Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods. In *Advances in Large Margin Classifiers*. MIT Press.
- Nina Rimskey, Nick Gabrieli, Julian Schulz, Meg Tong, Evan Hubinger, and Alexander Matt Turner. 2024. [Steering Llama 2 via contrastive activation addition](#). In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL)*.
- Aviv Slobodkin, Omer Goldman, Avi Caciularu, Ido Dagan, and Shauli Ravfogel. 2023. [The curious case of hallucinatory \(un\)answerability: Finding truths in the hidden states of over-confident large language models](#). In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*.
- Ben Snyder, Marius Moisescu, and Muhammad Bilal Zafar. 2024. [On early detection of hallucinations in factual question answering](#). In *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)*.

- Gaurang Sriramanan, Siddhant Bharti, Vinu Sankar Sadasivan, Shoumik Saha, Priyatham Kattakinda, and Soheil Feizi. 2024. [LLM-check: Investigating detection of hallucinations in large language models](#). In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 37, pages 34188–34216.
- Weihang Su, Changyue Wang, Qingyao Ai, Yiran Hu, Zhijing Wu, Yujia Zhou, and Yiqun Liu. 2024. [MIND: Unsupervised real-time hallucination detection based on the internal states of large language models](#). In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 14379–14391. Association for Computational Linguistics.
- Katherine Tian, Eric Mitchell, Allan Zhou, Archit Sharma, Rafael Rafailov, Huaxiu Yao, Chelsea Finn, and Christopher D. Manning. 2023. [Just ask for calibration: Strategies for eliciting calibrated confidence scores from language models fine-tuned with human feedback](#). In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*.
- Alexander Matt Turner, Lisa Thiergart, Gavin Leech, David Udell, Juan J. Vazquez, Ulisse Mini, and Monte MacDiarmid. 2023. [Steering language models with activation engineering](#).
- Raúl Vázquez, Timothee Mickus, Elaine Zosa, Teemu Vahtola, Jörg Tiedemann, Aman Sinha, Vincent Segonne, Fernando Sánchez-Vega, Alessandro Raganato, Jindřich Libovický, Jussi Karlgren, Shaoxiong Ji, Jindřich Helcl, Liane Guillou, Ona de Gibert, Jaione Bengoetxea, Joseph Attieh, and Marianna Apidianaki. 2025. [SemEval-2025 task 3: MU-SHROOM, the multilingual shared task on hallucinations and related observable overgeneration mistakes](#). In *Proceedings of the 19th International Workshop on Semantic Evaluation (SemEval-2025)*. Association for Computational Linguistics.
- Chris Wendler, Veniamin Veselovsky, Giovanni Monea, and Robert West. 2024. [Do llamas work in English? on the latent language of multilingual transformers](#). In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*.
- An Yang et al. 2025a. [Qwen3 technical report](#).
- Wannan Yang, Xinchu Qiu, Lei Yu, Yuchen Zhang, Aobo Yang, Narine Kokhlikyan, Nicola Cancedda, and Diego Garcia-Olano. 2025b. [Hallucination reduction with CASAL: Contrastive activation steering for amortized learning](#).
- Hanning Zhang, Shizhe Diao, Yong Lin, Yi R. Fung, Qing Lian, Xingyao Wang, Yangyi Chen, Heng Ji, and Tong Zhang. 2024. [R-Tuning: Instructing large language models to say ‘i don’t know’](#). In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 7113–7139.